

FULL REPORT

System Info

Product	Flow Simulation 2025 SP5.0. Build: 6875
Computer name	REMY
User name	26869
Processors	13th Gen Intel(R) Core(TM) i5-1340P
Memory	16068 MB / 4919 MB
Operating system	Windows 11 (or higher) (Version 10.0.26200)
CAD version	SOLIDWORKS 2025 SP5.0
CPU speed	1900 MHz

General Info

Model	Coletor.SLDASM
Project name	Project(1)
Project comment	
Project path	C:\Users\26869\Desktop\game\Solidwork\resource\flowmodeal\Flow Simulation\第一课\歧管\1
Units system	SI (m-kg-s)
Analysis type	Internal
Exclude cavities without flow conditions	On
Coordinate system	Global Coordinate System
Reference axis	X

INPUT DATA

Mesh

Global Mesh Settings

Automatic initial mesh: On

Result resolution level: 3

Advanced narrow channel refinement: Off

Geometry Resolution

Evaluation of minimum gap size: Automatic

Evaluation of minimum wall thickness: Automatic

Computational Domain

Size

X min	-0.146 m
X max	0.039 m
Y min	-0.037 m
Y max	0.062 m
Z min	-0.731 m
Z max	0.025 m
X size	0.184 m

Y size	0.099 m
Z size	0.756 m

Boundary Conditions

2D plane flow	None
At X min	Default
At X max	Default
At Y min	Default
At Y max	Default
At Z min	Default
At Z max	Default

Physical Features

Fluid Flow: On
 Conduction: Off
 Structural: Off
 Electromagnetics: Off
 Time dependent: Off
 Gravitational effects: Off
 Rotation: Off
 Flow type: Laminar and turbulent
 High Mach number flow: Off
 Humidity: Off
 Free surface: Off
 Default roughness: 0 micrometer
 Default wall conditions: Adiabatic wall

Initial Conditions

Thermodynamic parameters	Static Pressure: 101325.00 Pa Temperature: 293.20 K
Velocity parameters	Velocity vector Velocity in X direction: 0 m/s Velocity in Y direction: 0 m/s Velocity in Z direction: 0 m/s
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.001 m

Material Settings

Fluids

[Air](#)

Boundary Conditions

Inlet Volume Flow 1

Type	Inlet Volume Flow
Faces	Face<1>@LID10<1>
Coordinate system	Face Coordinate System
Reference axis	X

Flow parameters	Flow vectors direction: Normal to face Volume flow rate: 0.0500 m ³ /s Fully developed flow: No Inlet profile: 0
Thermodynamic parameters	Approximate pressure: 101325.00 Pa Temperature type: Temperature of initial components Temperature: 293.20 K
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.001 m
Boundary layer parameters	Boundary layer type: Turbulent

Static Pressure 2

Type	Static Pressure
Faces	Face<2>@LID17<1> Face<1>@LID14<1> Face<6>@LID6<1> Face<3>@LID11<1> Face<4>@LID1<1> Face<5>@LID7<1>
Coordinate system	Global Coordinate System
Reference axis	X
Thermodynamic parameters	Static pressure: 101325.00 Pa Temperature type: Temperature of initial components Temperature: 293.20 K
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.001 m
Boundary layer parameters	Boundary layer type: Turbulent

Goals

Global Goals

GG Average Static Pressure 1

Type	Global Goal
Goal type	Static Pressure
Calculate	Average value
Coordinate system	Global Coordinate System
Use in convergence	On

GG Average Total Pressure 2

Type	Global Goal
Goal type	Total Pressure
Calculate	Average value
Coordinate system	Global Coordinate System
Use in convergence	On

GG Average Dynamic Pressure 3

Type	Global Goal
Goal type	Dynamic Pressure
Calculate	Average value
Coordinate system	Global Coordinate System

Use in convergence	On
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Surface Goals

SG Volume Flow Rate 1

Type	Surface Goal
Goal type	Volume Flow Rate
Faces	Face<2>@LID17<1> Face<1>@LID14<1> Face<6>@LID6<1> Face<3>@LID11<1> Face<4>@LID1<1> Face<5>@LID7<1>
Coordinate system	Global Coordinate System
Use in convergence	On

Calculation Control Options

Finish Conditions

Finish Conditions	If one is satisfied
Maximum travels	4.000
Goals convergence	Analysis interval: 0.500

Solver Refinement

Refinement: Disabled

Solving

Results Saving

Save before refinement	On
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Advanced Control Options

Flow Freezing

Flow freezing strategy	Disabled
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RESULTS

General Info

Run at: REMY

Number of cores: 16

Iterations: 79

CPU time: 34 s

Log

Event	Iteration	Time
Mesh generation started	0	23:51:53 , May 08
Mesh generation normally finished	0	23:51:54 , May 08
Preparing data for calculation	0	23:51:54 , May 08
Calculation started	0	23:51:55 , May 08
Calculation has converged since the following criteria are	79	23:52:29 , May 08

satisfied:		
Goals are converged	79	
Calculation finished	79	23:52:35 , May 08

Warnings: A vortex crosses the pressure opening Boundary Condition : Static Pressure 2 ; Inlet flow/outlet flow=0.030366

Calculation Mesh

Basic Mesh Dimensions

Number of cells in X	8
Number of cells in Y	4
Number of cells in Z	34

Number Of Cells

Cells	12098
Fluid cells	12098
Irregular cells	0
Trimmed cells	0

Maximum refinement level: 2

Goals

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
GG Average Static Pressure 1	Pa	101356.21	100	1.18428842	0.218120943	On
GG Average Total Pressure 2	Pa	101415.33	100	1.18513657	0.911664053	On
GG Average Dynamic Pressure 3	Pa	59.09	100	0.804358965	0.755754787	On
SG Volume Flow Rate 1	m ³ /s	-0.0500	100	4.155498e-05	1.53390638e-05	On

Min/Max Table

Name	Minimum	Maximum
Density (Fluid) [kg/m ³]	1.20	1.21
Pressure [Pa]	101259.97	101460.14
Temperature [K]	293.15	293.29
Temperature (Fluid) [K]	293.15	293.29
Velocity [m/s]	0	15.178
Velocity (X) [m/s]	-14.094	4.681

Velocity (Y) [m/s]	-13.158	7.054
Velocity (Z) [m/s]	-14.521	4.764
Mach Number []	0	0.04
Velocity RRF [m/s]	0	15.178
Velocity RRF (X) [m/s]	-14.094	4.681
Velocity RRF (Y) [m/s]	-13.158	7.054
Velocity RRF (Z) [m/s]	-14.521	4.764
Vorticity [1/s]	0.33	5728.04
Dynamic Pressure [Pa]	0	138.62
Reference Pressure [Pa]	101325.00	101325.00
Relative Pressure [Pa]	-65.03	135.14
Shear Stress [Pa]	0	1.80
Bottleneck Number []	3.4943806e-07	1.0000000
Heat Transfer Coefficient [W/m^2/K]	0	0
ShortCut Number []	3.0435519e-07	1.0000000
Surface Heat Flux [W/m^2]	0	0
Surface Heat Flux (Convective) [W/m^2]	0	0
Total Enthalpy Flux [W/m^2]	-4625259.952	4730825.458
Acoustic Power [W/m^3]	0	1.412e-07
Acoustic Power Level [dB]	0	51.50

Engineering Database

Gases

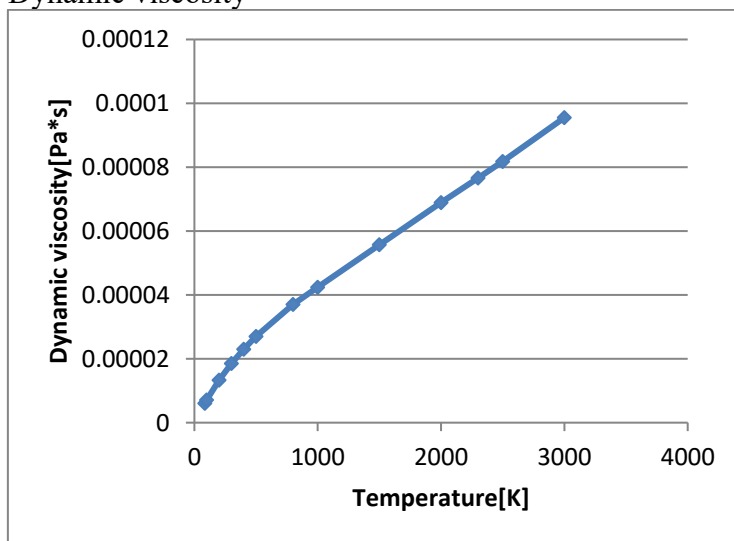
Air

Path: Gases Pre-Defined

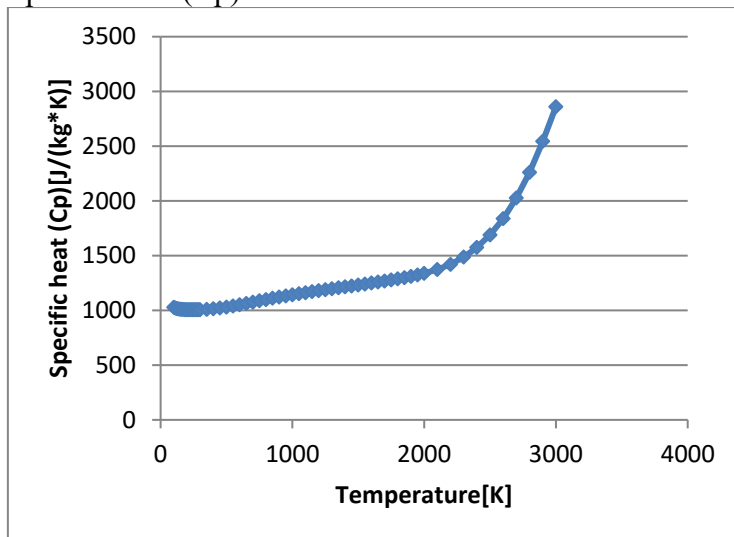
Specific heat ratio (C_p/C_v): 1.399

Molecular mass: 0.0290 kg/mol

Dynamic viscosity



Specific heat (Cp)



Thermal conductivity

